CAMCFormer: Cross-Attention and Multicorrelation Aided Transformer for Few-Shot Object Detection in Optical Remote Sensing Images

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***Abstract*— Few-shot object detection (FSOD) enables the detection of novel-class objects in remote sensing images (RSIs) with limited labeled samples. Although convolutional neural networks (CNNs) are commonly used for this task, they suffer from two inherent constraints. First, their limited local receptive field fails to capture global context within a single image and the relational dependencies between query and support images. Second, an additional feature alignment mechanism is typically required to bridge the gap between query and support images. To address these challenges, this work introduces a novel cross-attention and multicorrelation aided transformer (CAMCFormer) FSOD framework tailored for global feature representation and multicorrelation modeling in complex and large-scale RSIs. Specifically, a long-distance cross-attention module (LDCAM) is devised to capture dependencies between distant elements across query and support images at each feature extraction layer. This module facilitates the exchange of contextual information between images, resulting in more comprehensive feature representations and eliminating the need for separate feature alignment and fusion modules. Multicorrelation aided heads (MAHs) are constructed to enhance detection performance further to model various relational aspects, i.e., channel-correlation detection head (CCDH), spatial-correlation detection head (SCDH), and cross-attention detection head (CADH). These aided heads contribute to more robust and accurate classification and localization. Comprehensive experiments have been conducted, demonstrating the superiority of the proposed framework compared to several state-of-the-art detectors, highlighting its potential as an effective solution for FSOD in remote sensing scenarios.**

***Index Terms*— Few-shot learning (FSL), object detection, optical remote sensing images (RSIs), transformer.**

I. Introduction

With the development of deep learning, significant achievements have been made in the remote sensing field, such as object detection [1], [2], [3], [4], [5], image classification [6], semantic segmentation [7], [8], and so on [9], [10], [11]. Deep learning methods achieve superior performance but rely heavily on large datasets. However, acquiring large quantities of high-resolution remote sensing images (RSIs) is challenging due to uncontrollable dynamics, such as atmospheric conditions, sensor quality, satellite orbits, and data storage and transmission constraints. Moreover, annotating large volumes of high-resolution RSIs is labor-intensive and time-consuming, limiting the practical application of deep learning methods in RSI processing. Few-shot learning (FSL) [12], [13], [14] offers a promising research direction for understanding and monitoring the earth's natural and anthropogenic features. It enables rapid adaptation to new scenes and target categories by leveraging specialized training on existing datasets.

FSL can be categorized into meta-learning [15], [16], [17], transfer-learning [18], [19], [20], [21], and metric-learning [22], [23], [24], based on the type of knowledge learned from data. Meta-learning [25], [26], [27] focuses on enhancing a model's ability to learn how to learn. Transfer-learning [28], [29], [30] emphasizes leveraging domain-related knowledge by transferring insights from a pretrained model on a large dataset to a new, smaller dataset relevant to the target task. Metric-learning [31], [32] focuses on learning an effective distance metric to measure data similarity. Few-shot object detection (FSOD) applies FSL principles to object detection, focusing on designing tasks, model architectures, and training methodologies to achieve effective detection with limited data [33], [34], [35]. The research on FSOD in RSIs has focused on adapting existing FSOD frameworks [36], [37], [38], [39] to handle the unique challenges of RSIs, such as varying resolutions, sizes, extensive coverage areas, significant spatial distances, and complex scenes. Recent advancements aim to enhance the generalization ability of models by leveraging multimodal approaches [40], meta-learning [41], [42], and transfer learning [43], [44], [45] to detect objects with limited labeled samples in RSIs datasets effectively.

As shown in Table I, FSOD RSI methods typically integrate specialized modules into established object detection frameworks and training stages. The DETR framework and transformer-based feature extraction models are occasionally utilized in natural scenes. However, feature extraction is predominantly performed in remote sensing contexts using the Faster R-CNN [46] framework with convolutional neural

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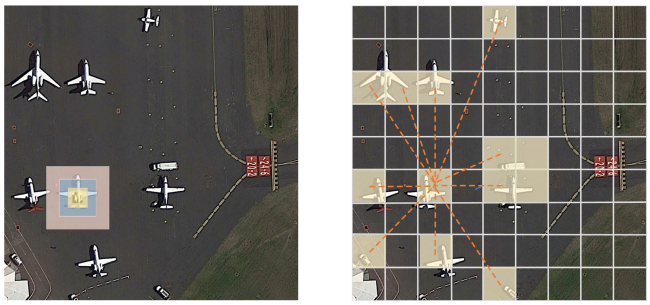
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TABLE I

OVERVIEW OF VARIOUS FSOD METHODS IN NATURAL AND REMOTE SENSING SCENARIOS. THE TABLE SUMMARIZES DIFFERENT METHODS, YEARS, OBJECT DETECTION FRAMEWORKS, BACKBONES, CATEGORIES, AND PROPRIETARY MODULES

Scenario	Methods	Years	Object Detection Framework	Backbone	Category	Proprietary Module
Natural	Meta R-CNN [55]	2019	Faster R-CNN	ResNet-101	Meta-learning	Predictor-head Remodeling Network (PRN)
	FsDetView [56]	2023	Faster R-CNN	ResNet-101	Meta-learning	Category-agnostic viewpoint estimation method
	FCT [57]	2022	Faster R-CNN	PVTv2	Meta-learning	Asymmetric-batched cross-attention
	FS-DETR [61]	2023	DETR	ResNet-101	Transfer-learning	Visual prompting
	Meta-DETR [59]	2022	DETR	ResNet-101	Meta-learning	Semantic Alignment Mechanism (SAM)
Remote Sensing	G-FSOD [44]	2023	Faster R-CNN	ResNet-101	Transfer-learning	Representation compensation module
	ST-FSOD [43]	2024	Faster R-CNN	ResNet-101	Transfer-learning	Two-branch region proposal networks
	AACE [45]	2023	Faster R-CNN	ResNet-101	Transfer-learning	Automatically learned aug-aware search module
	SAE-FSDet [62]	2024	Faster-RCNN	ResNet-101	Transfer-learning	Coarse-to-fine regression method
	TSF-RGR [58]	2023	Faster-RCNN	ResNet-101	Transfer-learning	Text semantic fusion relation graph reasoning
	TEMO [40]	2023	Faster-RCNN	ResNet-101	Transfer-learning	Text-modal knowledge extractor, Cross-modal assembly module
	FsCIT [63]	2023	Faster-RCNN	ResNet-101	Transfer-learning	Confidence-iou collaborative proposal filtration scheme
	TINet-FSOD [25]	2023	Faster R-CNN	ResNet-101	Meta-learning	Transformation-invariant network
	DPL-Net [42]	2023	Faster R-CNN	ResNet-101	Meta-learning	Fine-grained information fusion (FIF) module, Discriminative prototype learning (DPL) strategy
	FSODM [41]	2021	YOLOv3	DarkNet-53	Meta-learning	Feature reweighting module, Bounding box prediction module



(a) (b)

Fig. 1. Illustration of receptive fields and spatial relationships in CNNs and Transformers. (a) CNN-based methods: CNNs use convolutional kernels for local perception, extracting spatial features within a limited receptive field. Increasing layers or kernel size can expand this field, but CNNs focus on local features. (b) Transformer-based methods: Transformers capture long-range dependencies through self-attention, weighing the importance of different image parts and providing a global context for better understanding complex scenes.

network (CNN)-based networks. FSODM [41] is an exception, employing the YOLOv3 [47] framework with the DarkNet-53 backbone. CNNs like ResNet-101 [48], and DarkNet-53 are favored for their strong feature extraction capabilities. However, CNNs primarily focus on local regions through convolutional kernels. Despite increasing the number of layers or kernel size, CNNs primarily capture local features, limiting their ability to model long-range global contextual information. Equipped with self-attention mechanisms, transformers capture long-range dependencies and balance the importance of different image regions, providing global context, as shown in Fig. 1. This capability aids in understanding and interpreting the complex scenes and long-distance relationships typical in remote sensing imagery. Previous works have demonstrated that Transformers outperform traditional CNNs in various remote sensing tasks [49], such as classification [18], [50], object detection [51], semantic segmentation [52], and change detection [53], [54], due to their ability to model intricate spatial relationships more effectively. Therefore, our research focuses on enhancing the application of transformer structures within FSOD for RSIs.

Furthermore, we conduct a comprehensive analysis of the proprietary modules across different FSOD methods as shown in Table I, which reveals that the majority of innovations focus on the design of feature fusion modules [41], [42], [55], [56], [57], [58] and semantic alignment modules [25], [58], [59]. Specifically, meta-learning methods require the fusion of query and support features to find objects in query images from support images. However, significant discrepancies between query and support images necessitate an initial semantic alignment to ensure effective feature integration. Most related works based on CNNs necessitate additional design efforts for alignment and fusion to accomplish the FSOD task, typically focusing on aligning and fusing a single high-level semantic feature. CNNs can complicate information integration across different scales or channels, often necessitating additional design considerations, such as feature pyramid network (FPN) [60]. This limitation makes it challenging to establish associations among hundreds of extracted features.

FSOD model for RSIs uses a CNN-based detection framework almost exclusively. Still, its limited local receptive fields and the need for additional feature alignment mechanisms limit its ability to capture global context and relational dependencies between images. To tackle these challenges, this work introduces a novel framework called cross-attention and multicorrelation aided transformer (CAMCFormer) for FSOD, which is adept at modeling global relationships in complex and large-scale RSIs. Compared to CNN-based FSOD methods, the proposed method eliminates the need for complex feature fusion and semantic alignment modules. It builds

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cross-attention connections across queries and support among hundreds of feature extractors. Technically, the long-distance cross-attention module (LDCAM) is constructed to capture cross-image long-distance dependencies. This enables multiscale mutual learning during the feature extraction process, allowing the model to learn the importance of different parts across images under the cross-attention mechanism. Multicorrelation aided heads (MAHs) are designed to improve overall detection performance by reducing category confusion. This includes the channel-correlation detection head (CCDH), spatial-correlation detection head (SCDH), and cross-attention detection head (CADH), which perform parallel classification based on channel correlation, multispatial scale correlation, and cross-attention union correlation, respectively.

In summary, the main contributions of this article are as follows.

1) A transformer-based FSOD framework is proposed for optical RSIs, enhancing understanding of complex scenes and long-range features. The framework improves feature representation, accurately detecting small, densely packed, or irregularly shaped objects commonly found in remote sensing scenarios.
2) A LDCAM is introduced to model the importance of different parts across query and support images through the cross-attention mechanism, which enables mutual learning during the feature extraction process.
3) MAHs are devised to reduce category confusion and enhance detection performance, including CCDH, SCDH, and CADH.
4) Comprehensive experiments have been conducted over several widely adopted benchmark datasets, demonstrating the superiority of the proposed framework for FSOD.

The remaining parts of this article are organized as follows. Section II describes our proposed CAMCFormer method in detail. Comprehensive experimental results are shown in Section III. The conclusion of this article is in Section IV.

II. Methodology

A. Problem Formulation

FSOD extends the capability of a detector pretrained on base classes C_b with abundant data to novel classes C_n with limited training samples, where $C_n \cap C_b = \emptyset$. The datasets for the base and novel classes are denoted as $D_b = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{B}$ and $D_n = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{N}$ (with $N \ll B$, where B and N are the numbers of base class samples and novel class samples, respectively). Here, $\mathbf{x}_i$ and $\mathbf{x}_j$ represent RSIs containing objects, and $\mathbf{y}_i$ and $\mathbf{y}_j$ denote the object categories and their bounding box coordinates. In accordance with the meta-learning paradigm, a sequence of tasks $\mathcal{T} = \{\mathcal{T}_b, \mathcal{T}_n\}$ is generated from D_b and D_n, respectively. Each task consists of a query set $\boldsymbol{q}_i$ and a support set $\boldsymbol{s}_i$, where the support set includes c categories and k images per category, conforming to the C-way K-shot setup. During meta-training, the model learns to detect the query set based on the limited images in the support set, acquiring the knowledge of common structures. Then, transfer meta-knowledge from $\mathcal{T}_b = \{\boldsymbol{q}_i, \boldsymbol{s}_i\}$ to the tasks of novel classes $\mathcal{T}_n = \{\boldsymbol{q}_j, \boldsymbol{s}_j\}$ sampled from D_n, thereby improving detection performance on C_n. During the testing phase, all categories $C_t = C_b \cup C_n$ are evaluated simultaneously.

B. Overview of the Proposed Architecture

The overall architecture of our proposed CAMCFormer FSOD model is illustrated in Fig. 2, which employs the dependencies between support and query features across all feature extraction layers by leveraging the long-range modeling capabilities of transformers. The backbone employs a four-stage pyramid vision transformer (PVTv2) [64] architecture, which includes four patch embeddings and position encoders to generate multiscale patches. Technically, given a task $\mathcal{T} = \{\boldsymbol{q}, \boldsymbol{s}\}$ that included a single query image $\boldsymbol{q} \in \mathbb{R}^{1\times H\times W\times 3}$ and support images $\boldsymbol{s} \in \mathbb{R}^{c\times k\times h\times w\times 3}$ through patch embedding. In patch embedding, these images are split as nonoverlapping patches (4×4) and flattened, and then a linear projection is performed to obtain the embedded patches $\boldsymbol{P}_q^i \in \mathbb{R}^{1\times H_i W_i\times C_i}$, $\boldsymbol{P}_s^i \in \mathbb{R}^{c\times k\times h_i w_i\times C_i}$, $i \in \{1, 2, 3, 4\}$, where i is ith stage, $C_i \in \{64, 128, 320, 512\}$, and $H_i W_i = (H/4^i) * (W/4^i)$, $h_i w_i = (h/4^i) * (w/4^i)$. The embedded patches are progressively fed into position encoders to produce four-level query and support sequence of patches denoted as $\boldsymbol{Z}_q^i \in \mathbb{R}^{1\times H_i W_i\times C_i}$, $\boldsymbol{Z}_s^i \in \mathbb{R}^{c\times k\times h_i w_i\times C_i}$.

For all stages, we introduce the LDCAM, which leverages the long-range modeling capability of the cross-attention mechanism, enabling joint attention between the support and query images among each feature extraction channel. After stage 3, the support feature undergoes average pooling to obtain a unique support vector for each class. A meta-proposal generator processes the support vector and query features to acquire region proposals. All region proposals and support features are further extracted in stage 4 to obtain deeper features. These deep features are then input into the MAHs, where parallel and complementary classification is performed from multiple perspectives, including CCDH, SCDH, and CADH, ultimately completing the prediction.

C. Long-Distance Cross-Attention Module

Most previous methods require additional feature alignment [25], [58], [59] and fusion modules [41], [42], [55], [56], [57], [58]. Due to the challenges CNNs face in integrating multiscale and channel information, they can only perform a single alignment of high-dimensional semantic features, making it difficult to establish connections among hundreds of feature extraction channels. In contrast, the transformer architecture, composed of stacked self-attention mechanisms, allows for introducing a LDCAM. As shown in Fig. 3, this module facilitates long-distance relationship modeling between query and support features across each feature extraction channel, enabling sustained interactive attention. In the FSOD model for RSIs, the discriminative feature regions of the support set are crucial for identifying the query set objects. Sustained interactive attention across query and support features can effectively emphasize regions that co-occur and eliminate feature alignment and fusion modules.

Specifically, embedded patches $\boldsymbol{P}_q^i$ and $\boldsymbol{P}_s^i$ must first be augmented with positional encoding $\boldsymbol{E}_q^{\text{pos}} \in \mathbb{R}^{H_i W_i\times C_i}$, $\boldsymbol{E}_s^{\text{pos}} \in$

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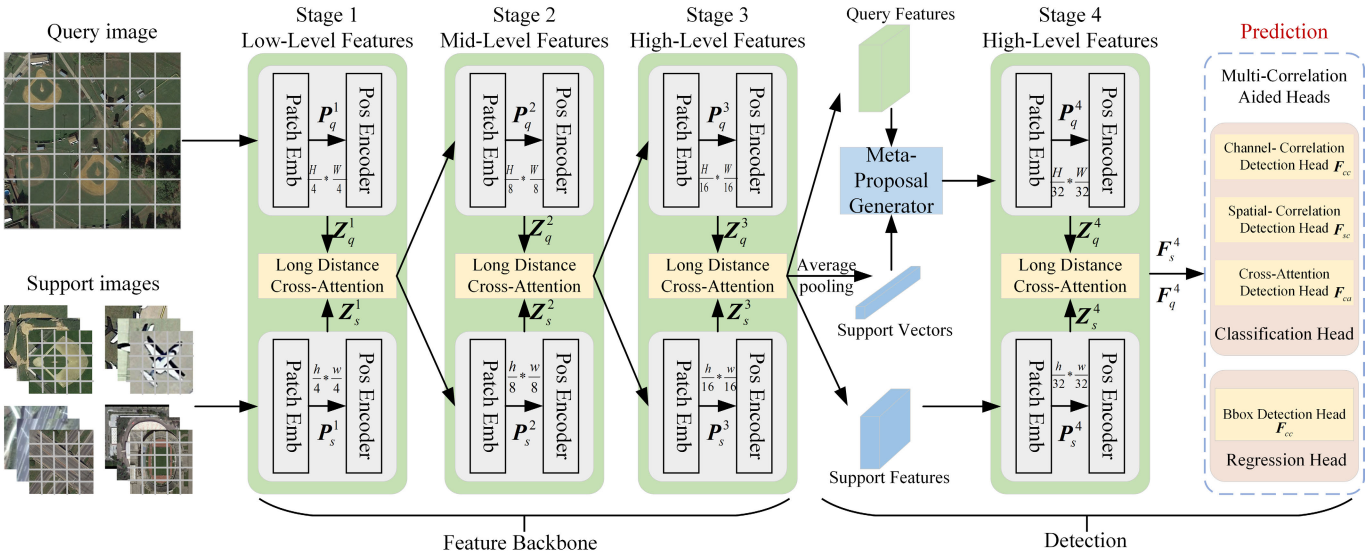


Fig. 2. Overview of the proposed CAMCFormer architecture, which consists of four stages: low-level, mid-level, and high-level feature extractors integrated with the LDCAM, a meta-proposal generator, high-level detection feature extraction, and MAHs. To obtain pyramid features in the transformer framework, the features from each stage are divided into nonoverlapping embedding blocks. Subsequently, LDCAM is applied at each stage to achieve cross-image attention between support and query images. After the high-level feature extraction stage, support features are averaged to obtain a unique support vector for each class. The meta-proposal generator processes the support vectors and query features to generate region proposals. Finally, MAHs perform complementary classification from multiple perspectives and single regression for detection. The classification heads include the CCDH, SCDH, and CADH, completing the final prediction.

$\mathbb{R}^{h_i w_i \times C_i}$ and set encoding $\boldsymbol{E}^{\text{set}} \in \mathbb{R}^{2*C_i}$ to distinguish the patch positions in subsequent LDCAM. The formula is as follows:

$$\begin{aligned} \boldsymbol{Z}_q^i &= \boldsymbol{P}_q^i + \boldsymbol{E}_q^{\text{pos}} + \boldsymbol{E}^{\text{set}}[0] \\ \boldsymbol{Z}_s^i &= \boldsymbol{P}_s^i + \boldsymbol{E}_s^{\text{pos}} + \boldsymbol{E}^{\text{set}}[1]. \end{aligned} \tag{1}$$

The sequence of patches $\boldsymbol{Z}_q^i, \boldsymbol{Z}_s^i$ is projected onto three learnable matrices $\boldsymbol{W}_Q, \boldsymbol{W}_K, and \boldsymbol{W}_V$ to obtain $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, following the Q–K–V attention in transformer:

$$\begin{aligned} \boldsymbol{Q}_q^m &= \boldsymbol{Z}_q^i \boldsymbol{W}_Q^m, \quad \boldsymbol{Q}_s^m = \boldsymbol{Z}_s^i \boldsymbol{W}_Q^m \\ \boldsymbol{K}_q^m &= \text{SR}\left(\boldsymbol{Z}_q^i\right) \boldsymbol{W}_K^m, \quad \boldsymbol{K}_s^m = \text{SR}\left(\boldsymbol{Z}_s^i\right) \boldsymbol{W}_K^m \\ \boldsymbol{V}_q^m &= \text{SR}\left(\boldsymbol{Z}_q^i\right) \boldsymbol{W}_V^m, \quad \boldsymbol{V}_s^m = \text{SR}\left(\boldsymbol{Z}_s^i\right) \boldsymbol{W}_V^m \end{aligned} \tag{2}$$

where $m = \{1, 2, \ldots, M\}$ represents mth head of multihead cross-attention, M is the number of heads, and $\boldsymbol{W}_Q^m, \boldsymbol{W}_K^m$, and $\boldsymbol{W}_V^m \in \mathbb{R}^{C_i * (C_i/M)}$ are the learnable weights of the linear projection, which are shared between query and support features. To reduce the computational complexity of the attention, linear spatial reduction SR() is performed on $\boldsymbol{Z}_q^i$ and $\boldsymbol{Z}_s^i$ according to the linear spatial reduction attention (SRA) operation of PVTv2 [64].

The Q–K–V sequences of query $\boldsymbol{Q}_q^m, \boldsymbol{K}_q^m, and \boldsymbol{V}_q^m$ and support $\boldsymbol{Q}_s^m, \boldsymbol{K}_s^m, and \boldsymbol{V}_s^m$ are fed into LDCAM, which maintains the multihead format

$$\begin{aligned} \boldsymbol{Z}_q^{i\prime} &= \text{LDCAM}\left(\boldsymbol{Q}_q, \boldsymbol{K}_q, \boldsymbol{V}_q\right) \\ &= \text{Concat}\left(\text{head}_q^1, \text{head}_q^2, \ldots, \text{head}_q^M\right) \boldsymbol{W}^O \\ \boldsymbol{Z}_s^{i\prime} &= \text{LDCAM}\left(\boldsymbol{Q}_s, \boldsymbol{K}_s, \boldsymbol{V}_s\right) \\ &= \text{Concat}\left(\text{head}_s^1, \text{head}_s^2, \ldots, \text{head}_s^M\right) \boldsymbol{W}^O \end{aligned} \tag{3}$$

where $\boldsymbol{W}^O$ represents the weight of the projection back to the original feature space, shared with the query and support features. The process of m-head $\text{head}_q^m, \text{head}_s^m$ in the long-distance cross-attention is formulated as

$$\begin{aligned} \boldsymbol{Z}_q^m &= \text{head}_q^m = \text{Cross_Atten}\left(\boldsymbol{Q}_q^m, \boldsymbol{K}_q^m, \boldsymbol{V}_q^m\right) \\ \boldsymbol{Z}_s^m &= \text{head}_s^m = \text{Cross_Atten}\left(\boldsymbol{Q}_s^m, \boldsymbol{K}_s^m, \boldsymbol{V}_s^m\right) \end{aligned} \tag{4}$$

where Cross_Atten() contains a series of long-distance cross-attention operations. First, the cross dot-product $\text{Att}_q, \text{Att}_s$ performs a matrix multiplication MatMul() between the query tensor $\boldsymbol{Q}_q^m$ and the concatenated key tensors to measure the similarity. The calculation process is as follows:

$$\begin{aligned} \text{Att}_q &= \text{MatMul}\left(\boldsymbol{Q}_q^m, \text{Concat}\left(\boldsymbol{K}_q^m, \frac{1}{k}\sum_k \boldsymbol{K}_s^m\right)\right) \\ \text{Att}_s &= \text{MatMul}\left(\boldsymbol{Q}_s^m, \text{Concat}\left(\boldsymbol{K}_s^m, \text{Rep}\left(\boldsymbol{K}_q^m, k\right)\right)\right) \end{aligned} \tag{5}$$

where $(1/k)\sum_k \boldsymbol{K}_s^m$ represents the averaged support key tensors for k-shot and $\text{Rep}(\boldsymbol{K}_q^m, k)$ denotes the $\boldsymbol{K}_q^m$ is replicated k times. This ensures that the query and support have consistent dimensions, facilitating efficient computation. From (5), it can be observed that the LDCAM module calculates the cross dot-product by simultaneously considering query self-attention and cross-attention between the query and support patches.

Next, the attention scores $\text{Att}_q{}', \text{Att}_s{}'$ are calculated using the softmax function followed by the standard self-attention operations. The process is formally expressed by the following equation:

$$\begin{aligned} \text{Att}_q{}' &= \text{Softmax}\left(\text{Mask}\left(\text{Scale}\left(\text{Att}_q\right)\right)\right) \\ \text{Att}_s{}' &= \text{Softmax}(\text{Mask}(\text{Scale}(\text{Att}_s))) \end{aligned} \tag{6}$$

where Scale() is a scaling factor, typically the inverse square root of the dimensionality of the key vectors to stabilize the gradients during training. To compute the weighted sum of the

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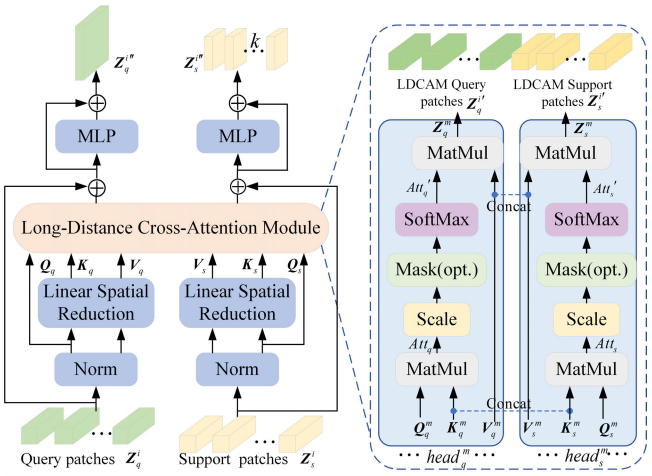


Fig. 3. Illustration of LDCAM. The detailed process of the multihead cross-attention step is illustrated, focusing on the mth head within the LDCAM.

value vectors, the matrix multiplication combines the attention scores $\mathrm{Att}_q{}'$ and $\mathrm{Att}_s{}'$ with the concatenated value vectors, shown in the following equation:

$$\begin{aligned} \boldsymbol{Z}_q^m &= \mathrm{MatMul}\left(\mathrm{Att}_q{}', \mathrm{Concat}\left(\boldsymbol{V}_q^m, \frac{1}{k}\sum_k \boldsymbol{V}_s^m\right)\right) \\ \boldsymbol{Z}_s^m &= \mathrm{MatMul}\left(\mathrm{Att}_s{}', \mathrm{Concat}\left(\boldsymbol{V}_s^m, \mathrm{Rep}\left(\boldsymbol{V}_q^m, k\right)\right)\right). \end{aligned} \tag{7}$$

$\boldsymbol{Z}_q^m$ and $\boldsymbol{Z}_s^m$ are the outputs of mth head of LDCAM. According to (7), the weighted sum of the value vectors for the query and support patches integrates their respective information.

Then, the feed-forward network is applied to LDCAM patches with stronger feature representations as following:

$$\begin{aligned} \boldsymbol{Z}_q^{i\,\prime\prime} &= \mathrm{MLP}\left(\mathrm{LN}\left(\boldsymbol{Z}_q^{i\,\prime}\right)\right) + \boldsymbol{Z}_q^{i\,\prime} \\ \boldsymbol{Z}_s^{i\,\prime\prime} &= \mathrm{MLP}\left(\mathrm{LN}\left(\boldsymbol{Z}_s^{i\,\prime}\right)\right) + \boldsymbol{Z}_s^{i\,\prime} \end{aligned} \tag{8}$$

where MLP() is multilayer perceptrons and LN() is layer normalization operation.

As a result, all feature extraction channels in the backbone incorporate interactive attention between the query and support, enabling continuous mutual learning without additional alignment and fusion modules. After features backbone stage 3, the support features are averaged to generate a single support vector for each class. The meta-proposal generator uses this support vector as a convolution kernel to convolve with the query features and generate proposals [16]. After detection stage 4, RoIAlign [65] is employed to extract the initial RoI features $\boldsymbol{F}_q^4 \in \mathbb{R}^{128\times H_4 W_4 \times C_4}$ and support features $\boldsymbol{F}_s^4 \in \mathbb{R}^{c\times k\times h_4 w_4 \times C_4}$ (128 is the default proposal parameter, and $H_4 = W_4 = h_4 = w_4 = 7$ is the default spatial size after RoIAlign) are similarly processed.

The model used in this article primarily follows the PVTv2-B2-Li setup, with all self-attention modules replaced by LDCAM. According to the model design, the ith stage contains depth_i stacked LDCAM modules. As the network layers deepen, the channel dimensions increase while the spatial resolution decreases. Consequently, Stage 3 incurs the highest computational cost. Deeper networks imply more substantial expressive power but require more computational resources and memory overhead. Under the hardware conditions used in this article, the PVTv2-B2-Li setup was employed. With more powerful hardware resources, it may be possible to explore deeper network configurations. The design particulars of the model are presented in Table II.

TABLE II
CONFIGURATION OF CAMCFORMER ACROSS DIFFERENT STAGES AND VARIANTS

	Output size	Layer name	CAMCFormer(PVTv2) B2-Li
Stage 1	$\frac{H}{4}\times\frac{W}{4}$, $\frac{h}{4}\times\frac{w}{4}$	Patch Embedding	$S_1 = 4$
		Position Encoder	$C_1 = 64$
		LDCAM	$P_1 = 7$ $N_1 = 1$ $E_1 = 8$ $depth_1 = 3$
Stage 2	$\frac{H}{8}\times\frac{W}{8}$, $\frac{h}{8}\times\frac{w}{8}$	Patch Embedding	$S_2 = 2$
		Position Encoder	$C_2 = 128$
		LDCAM	$P_2 = 7$ $N_2 = 2$ $E_2 = 8$ $depth_2 = 3$
Stage 3	$\frac{H}{16}\times\frac{W}{16}$, $\frac{h}{16}\times\frac{w}{16}$	Patch Embedding	$S_3 = 2$
		Position Encoder	$C_3 = 320$
		LDCAM	$P_3 = 7$ $N_3 = 5$ $E_3 = 4$ $depth_3 = 6$
Stage 4	$\frac{H}{32}\times\frac{W}{32}$, $\frac{h}{32}\times\frac{w}{32}$	Patch Embedding	$S_4 = 2$
		Position Encoder	$C_4 = 512$
		LDCAM	$P_4 = 7$ $N_4 = 8$ $E_4 = 4$ $depth_4 = 3$

1) S_i: Stride of the overlapping patch embedding
2) C_i: Number of channels of output.
3) R_i: Reduction ratio of the SRA.
4) P_i: Adaptive average pooling size of the linear SRA.
5) N_i: Number of heads of the efficient self-attention.
6) E_i: Expansion ratio of the feed-forward layer.
7) $depth_i$: Number of LDCAM.

D. Multicorrelation Aided Heads

To effectively handle complex remote sensing scenarios, we propose MAHs, which enable the model to gather complementary insights from various feature representations, enhancing the comprehensive feature analysis capability and improving the model's overall detection performance. CCDH focuses on the relationships between support and query feature channels, SCDH captures multiperspective spatial relationships between query and support features, and CADH extracts relevant information from the two feature maps, integrating crucial data from the support feature map into the query feature

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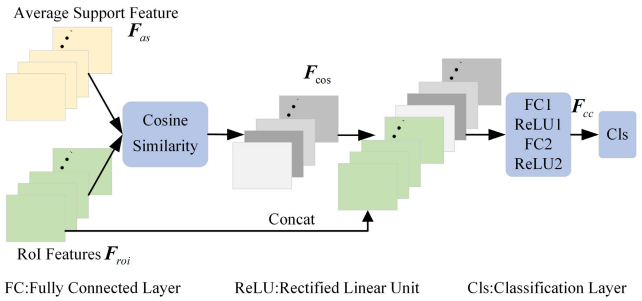


Fig. 4. Illustration of CCDH.

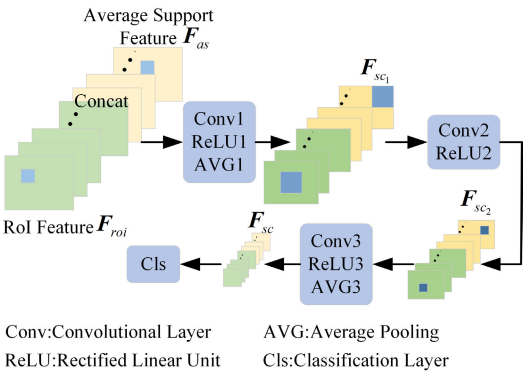


Fig. 5. Illustration of SCDH.

map. This multiangle approach ensures the model captures extensive information, thus increasing its ability to detect objects in complex RSIs accurately.

1) Channel-Correlation Detection Head: As shown in Fig. 4, the CCDH computes the overall cosine similarity between each channel of RoI features $\boldsymbol{F}_{\text{roi}} \in \mathbb{R}^{128\times7\times7\times C_4}$ and the average support feature $\boldsymbol{F}_{\text{as}} = (1/k)\sum_1^k \boldsymbol{F}_s^4, \boldsymbol{F}_{\text{as}} \in \mathbb{R}^{1\times7\times7\times C_4}$. The cosine similarity $\boldsymbol{F}_{\cos}$ reflects the comprehensive similarity for each channel. The formula is expressed as follows:

$$\boldsymbol{F}_{\cos} = \cos(\boldsymbol{F}_{\text{roi}}, \boldsymbol{F}_{\text{as}}) = \frac{\boldsymbol{F}_{\text{roi}} \cdot \boldsymbol{F}_{\text{as}}}{\|\boldsymbol{F}_{\text{roi}}\|\|\boldsymbol{F}_{\text{as}}\|}. \tag{9}$$

The $\boldsymbol{F}_{\text{roi}}$ and cosine similarity $\boldsymbol{F}_{\cos}$ are concatenated and passed through two fully connected layers followed by ReLU activation layers to obtain channel-correlation features $\boldsymbol{F}_{\text{cc}}$. Finally, the channel-correlation classifier performs classification based on the channel-correlation features, shown in the following equation:

$$\begin{aligned} \boldsymbol{F}_{\text{cc}} &= \text{fc}(\text{concat}(\boldsymbol{F}_{\text{roi}}, \boldsymbol{F}_{\cos})) \\ f_{\text{cc}} &= \text{cls}(\boldsymbol{F}_{\text{cc}}) \end{aligned} \tag{10}$$

where fc() represents two fully connected layers and ReLU activation layers, f_{cc} is channel-correlation classification score, and cls() represents a classification layer.

2) Spatial-Correlation Detection Head: As shown in Fig. 5, the SCDH extracts spatial-correlation features $\boldsymbol{F}_{\text{sc}}$ between $\boldsymbol{F}_{\text{roi}}$ and $\boldsymbol{F}_{\text{as}}$ at different scales by connecting the RoI features with the averaged support features. Subsequently, a series of convolutional layers with kernels of varying sizes, along with pooling layers, are applied to capture multiscale spatial relationships between the features. The process can be mathematically represented as follows:

$$\boldsymbol{F}_{\text{sc}_1} = \text{AVG1}(\text{ReLU1}(\text{conv1}(\text{concat}(\boldsymbol{F}_{\text{roi}}, \boldsymbol{F}_{\text{as}})))) \tag{11}$$

where conv1() is a convolution operation with a kernel size of 1, AVG1() denotes an average pooling operation with a kernel size of 3, and $\boldsymbol{F}_{\text{sc}_1} \in \mathbb{R}^{128\times5\times5\times2C_4}$

$$\boldsymbol{F}_{\text{sc}_2} = \text{ReLU2}\big(\text{conv2}\big(\boldsymbol{F}_{\text{sc}_1}\big)\big) \tag{12}$$

where conv2() is a convolution operation with a kernel size of 3 and $\boldsymbol{F}_{\text{sc}_2} \in \mathbb{R}^{128\times3\times3\times(C_4/2)}$. The spatial-correlation classification score f_{sc} is expressed as follows:

$$\begin{aligned} \boldsymbol{F}_{\text{sc}} &= \text{AVG3}\big(\text{ReLU3}\big(\text{conv3}\big(\boldsymbol{F}_{\text{sc}_2}\big)\big)\big) \\ f_{\text{sc}} &= \text{cls}(\boldsymbol{F}_{\text{sc}}) \end{aligned} \tag{13}$$

where the kernel size of conv3() also is 1, the kernel size of AVG3() is 3, and $\boldsymbol{F}_{\text{sc}} \in \mathbb{R}^{128\times1\times1\times C_4}$.

3) Cross-Attention Detection Head: LDCAM and CADH involve similar operations, and there are also significant differences. The primary distinction between LDCAM and CADH lies in their approach to cross-attention: LDCAM is bidirectional, while CADH is unidirectional. This difference arises from their positions and roles within the models. LDCAM is situated within the backbone feature extraction process, where the dual-branch structure of the support and query sets allows for bidirectional cross-attention, eliminating the need for additional fusion and alignment modules, and enabling continuous mutual attention across multiple stacked feature extraction layers. The LDCAM focuses on the long-distance relationship between the query and support features, while the CADH emphasizes the relevant regions of the average support feature in RoI features. As shown in Fig. 6, the CADH calculates cross-attention scores that focus on the regions within $\boldsymbol{F}_{\text{roi}}$ relevant to the $\boldsymbol{F}_{\text{as}}$, and suppress irrelevant ones to enhance overall performance in FSOD.

Specifically, the Q–K–V sequence is first generated according to the attention mechanism

$$\boldsymbol{Q}_{\text{roi}} = \boldsymbol{F}_{\text{roi}}\boldsymbol{W}_Q^{\text{roi}}, \quad \boldsymbol{K}_{\text{as}} = \boldsymbol{F}_{\text{as}}\boldsymbol{W}_K^{\text{as}}, \quad \boldsymbol{V}_{\text{roi}} = \boldsymbol{F}_{\text{roi}}\boldsymbol{W}_S^{\text{roi}} \tag{14}$$

where $\boldsymbol{Q}_{\text{roi}}$, $\boldsymbol{V}_{\text{roi}}$ are derived from $\boldsymbol{F}_{\text{roi}}$, and $\boldsymbol{K}_{\text{as}}$ is produced from $\boldsymbol{F}_{\text{as}}$. The scaled dot-product attention score of RoI Att_{roi} is calculated from $\boldsymbol{Q}_{\text{roi}}$ and $\boldsymbol{K}_{\text{as}}$

$$\text{Att}_{\text{roi}} = \text{MatMul}\big(\boldsymbol{Q}_{\text{roi}}, \text{rep}(\boldsymbol{K}_{\text{as}}, 128)\big) \tag{15}$$

where $\text{rep}(\boldsymbol{K}_{\text{as}}, 128)$ replicates $\boldsymbol{K}_{\text{as}}$ 128 times to align with $\boldsymbol{Q}_{\text{roi}}$. The calculation of the cross-attention score $\text{Att}_{\text{roi}}{}'$ follows the conventional method, as shown:

$$\begin{aligned} \text{Att}_{\text{roi}}{}' &= \text{Softmax}(\text{Mask}(\text{Scale}(\text{Att}_{\text{roi}}))) \\ \boldsymbol{F}_{\text{ca}} &= \text{MatMul}\big(\text{Att}_{\text{roi}}{}', \boldsymbol{V}_{\text{roi}}\big) \end{aligned} \tag{16}$$

where cross-attention features $\boldsymbol{F}_{\text{ca}} \in \mathbb{R}^{128\times7\times7\times(C_4/2)}$. Finally, convolution, activation, and average pooling are used to adjust the $\boldsymbol{F}_{\text{ca}}$ dimensions for classification. The equations are as follows:

$$f_{\text{ca}} = \text{cls}(\text{AVG4}(\text{ReLU4}(\text{conv4}(\boldsymbol{F}_{\text{ca}})))) \tag{17}$$

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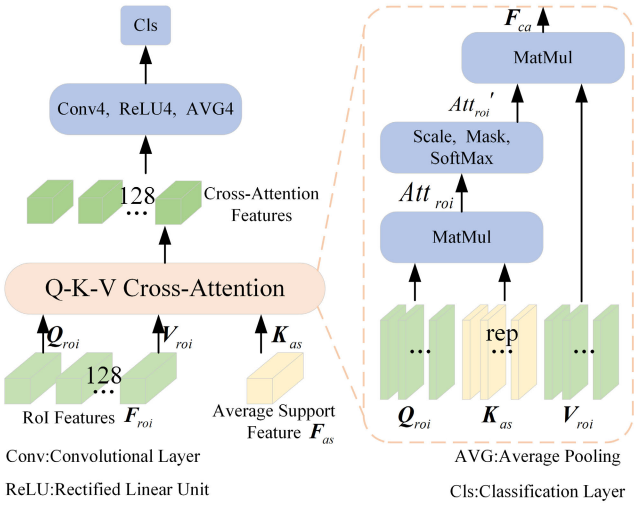


Fig. 6. Illustration of CADH.

where f_{ca} is cross-attention classification score, the kernel size of conv4() is 1, and the kernel size of AVG4() is 7.

The total classification score f_{total} is the sum of the classification scores from MHAs. Since the regression task focuses on estimating continuous bounding box coordinates rather than classifying discrete categories, it is less complex than the classification task. Using a single regression head is computationally efficient and sufficient for learning spatial relationships and predicting the precise locations of objects in an image. Therefore, the box prediction f_{box} utilizes a single bounding box regression head. The regression features using $\boldsymbol{F}_{cc}$ were derived from the experimental results and the correlation analysis

$$\begin{aligned} f_{total} &= f_{cc} + f_{sc} + f_{ca} \\ f_{box} &= \text{reg}(\boldsymbol{F}_{cc}) \end{aligned} \tag{18}$$

where reg() denotes regression function.

III. Experiments and Results

This section presents the experimental validation of the proposed algorithm on two public benchmark datasets (NWPU VHR-10 [71] and DIOR [72]) for RSI object detection. It encompasses datasets, experimental configurations, comparative experiments, ablation studies, and analysis of model architectures.

A. Datasets

The NWPU VHR-10 [71] dataset is a high-resolution remote sensing imagery dataset widely used for object detection tasks in remote sensing. It is introduced to provide a comprehensive set of annotated images for evaluating and benchmarking object detection algorithms in the context of very high-resolution (VHR) aerial imagery. The dataset comprises 800 images, each with a 1000 × 1000 pixels resolution. It contains 3651 labeled objects distributed across ten categories.

The DIOR [72] dataset consists of 23 463 images with a resolution of approximately 800 × 800 pixels, containing 192 472 annotated instances across 20 object categories. These categories include airplanes, ships, vehicles, and other objects commonly found in aerial imagery. The images in the DIOR dataset are collected from various sensors and platforms, providing a rich diversity in geographical locations, seasons, weather conditions, and imaging angles. This diversity makes the DIOR dataset particularly valuable for developing and evaluating robust object detection algorithms that generalize well across different situations and environments.

B. Experimental Settings

Two datasets, NWPU VHR-10 and DIOR, are partitioned into the base and novel subsets based on their classes to evaluate the performance of few-shot object detectors in remote sensing scenarios. As shown in Table III, the NWPU VHR-10 dataset is divided into three novel classes, with the remaining classes as base classes. The DIOR dataset is designated five novel classes and considers the remaining classes as base classes. All experiments' query images are sized at the original resolution, corresponding to the images from the DIOR and NWPU datasets. Support images containing only one target are cropped based on the ground truth bounding boxes from the DIOR and NWPU datasets and uniformly resized to 224 × 224 pixels.

Both the support and query sets consist of base classes during the meta-training stage. In each epoch, the support set comprises randomly sampled few-shot samples of C-way K-shot. During the meta-testing stage, the support and query sets are derived from a small balanced dataset containing all classes, including both base and novel classes. The support set for base classes consists of randomly sampled K-shot samples, while for novel classes, it consists of the only available K-shot samples. The value of K is set to 3, 5, 10, and 20, respectively.

To balance the effectiveness and efficiency, we adopt PVTv2-B2-Li as the backbone and initialize it with the pre-trained parameters. The anchor ratios are 0.25, 0.5, 1, 2, and 4. The models are optimized using Adam optimizer for 20 epochs with a base learning rate of 0.0002 which will decay to 1/10 every 10 epochs. Additionally, all experiments are conducted using PyTorch with an NVIDIA GeForce RTX 3090 GPU (24 GB memory).

C. Comparisons With State-of-the-Art Methods

To conduct a fair comparison with the previously developed models, all models are trained on the NWPU VHR-10 and DIOR datasets, respectively, including Meta R-CNN [55], FsDetView [56], DAnA [66], FSCE [69], MFDC [68], MLII-FSOD [70], and FCT [57]. All model results are obtained from their publicly available source. To validate the effectiveness of the proposed method, we compared it with several state-of-the-art methods. All experiments are conducted with identical hyperparameters, such as the number of iterations, decay rate, and anchor size, to ensure a fair comparison. To minimize the impact of varying support images on the experimental results, different random seeds are used to select support images in each experiment, which are repeated ten times to obtain average values, thereby ensuring robust and reliable results. Unlike

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TABLE III
DETAILED INFORMATION, NOVEL, AND BASE CLASSES SPECIFIED FOR THE NWPU VHR-10 AND DIOR DATASETS IN THE EXPERIMENTAL SETUP

Dataset	Classes	Images	Instances	Resolution	Novel Classes	Base Classes
NWPU VHR-10	10	800	3651	1000 × 1000	airplane, tennis court, baseball diamond	ship, storage tank, harbor, bridge, vehicle, ground track field
DIOR	20	23463	192472	800 × 800	windmill, airplane, tennis court, train station, baseball field	dam, stadium, ground track field, basketball court, golf field, storage tank, airport, vehicle, bridge, harbor, ship, expressway service area, expressway toll station, overpass, chimney

TABLE IV
QUANTITATIVE EXPERIMENTAL RESULTS OF THE PROPOSED METHOD AND EXISTING SOTA FSOD METHODS ON NWPU VHR-10 DATASET. BOLD FONT INDICATES THE BEST RESULT IN THE GROUP. SYMBOL '-' REPRESENTS UNREPORTED RESULTS IN THE ORIGINAL WORK. THE RESULTS ARE REPORTED BY AVERAGING OVER 10 RUNS

Methods	3-Shot				5-Shot				10-Shot				20-Shot			
	AL	BD	TC	nAP	AL	BD	TC	nAP	AL	BD	TC	nAP	AL	BD	TC	nAP
Meta R-CNN [55]	6.9	7.2	5.8	6.6	12.3	21.7	1.9	12.0	11.5	22.4	12.5	15.5	13.6	24.6	16.8	18.3
FsDetView [56]	9.1	19.7	6.8	11.9	14.3	26.8	10.8	17.3	16.1	20.1	16.4	17.5	18.8	30.2	12.4	20.5
DAnA [66]	16.2	10.3	13.5	13.3	15.6	14.5	12.9	14.3	27.7	24.4	19.6	23.9	28.8	27.0	23.0	26.3
DCFS [67]	18.4	38.2	17.3	25.0	34.6	70.2	18.5	41.1	52.8	80.7	36.2	56.6	44.5	78.6	34.4	52.5
MFDC [68]	31.8	36.9	17.7	28.8	50.2	57.3	21.6	43.0	62.4	69.1	26.8	52.7	55.4	71.7	34.4	53.8
FSCE [69]	29.1	51.6	25.0	35.2	42.8	66.0	26.3	45.1	60.7	76.8	33.0	56.8	72.3	81.9	42.9	65.7
MLII-FSOD [70]	42.3	57.5	30.2	43.3	52.4	71.3	47.7	57.1	74.2	72.1	52.0	66.1	**88.7**	72.2	53.3	71.4
FCT [57]	49.4	37.9	42.2	43.2	53.3	53.6	47.8	51.6	58.8	88.0	57.3	68.1	76.8	90.4	65.1	77.4
CAMCFormer(ours)	**51.5**	**57.8**	**53.1**	**54.2**	**58.9**	**72.1**	**49.1**	**60.0**	**79.8**	**89.5**	**62.2**	**77.2**	83.4	**90.6**	**66.3**	**80.1**

Abbreviation of novel classes - AL: airplane; BD: baseball diamond; TC: tennis court.

the performance evaluation of conventional object detection algorithms, in addition to the overall evaluation metric mAP, FSOD algorithms also distinguish between base class average precision (bAP) and novel class average precision (nAP) to evaluate the detection performance of base classes and novel classes, respectively.

1) Experiments on NWPU VHR-10: Table IV shows the experiment results on the NWPU VHR-10 dataset, and the novel classes airplane, baseball diamond, and tennis court are abbreviated as AL, BD, and TC, respectively. Among the compared methods, both the FCT [57], which is based on a transformer architecture, and the recently designed MLII-FSOD [70] for remote sensing achieved commendable results. The FCT leverages transformers' powerful feature extraction capabilities, demonstrating excellent performance in detecting small objects against complex backgrounds. Meanwhile, the MLII-FSOD enhances the detection accuracy of rare categories by incorporating a multilevel information integration mechanism. From the results, we can observe that our proposed method consistently outperformed the others in terms of single-category and overall average precision in novel classes across nearly all shot settings. Specifically, the average precision in novel classes improve 25.2%, 5.1%, 13.4%, and 3.5% in the 3-shot, 5-shot, 10-shot, and 20-shot settings Notably, our method exhibited significant advantages in low-shot scenarios, where it was more effective in learning key features of new categories, thereby achieving higher detection accuracy. Additionally, our method demonstrated superior robustness and generalization for challenging categories such as BD and TC, further validating its effectiveness.

As illustrated in Fig. 7, we present the detection results of the proposed CAMCFormer method, compared to a naive transformer framework (without the LDCAM) with MAHs and a CNN-based ResNet-101, all using the same backbone architecture. The results demonstrate that the CAMCFormer performs best across these novel classes. Specifically, the CAMCFormer exhibits superior detection accuracy and robustness, with fewer missed detections and misclassifications. In contrast, the naive transformer framework shows a noticeable decline in performance, particularly in detecting small and cluttered objects. This highlights the importance of the LDCAM in enhancing the model's ability to capture fine-grained features and contextual information. Similarly, the CNN-based ResNet-101 also struggles with more frequent missed detections and misclassifications, especially in complex scenes where the background is highly variable. The results show that the CAMCFormer achieves the best overall performance. In contrast, the naive transformer framework and the CNN-based ResNet-101 suffer from more missed detections and misclassifications.

2) Experiments on DIOR: The proposed model is compared with state-of-the-art methods on the DIOR dataset. The experimental results are demonstrated in Table V. Specifically, the average precision in novel classes improves by 18.1%, 10.8%, 12.2%, and 16.1% in the 3-shot, 5-shot, 10-shot, and 20-shot settings, respectively, indicating its effectiveness across different scenarios. The proposed method achieved optimal performance across novel classes. The visualized results are shown in Fig. 8, which reveal that the windmill category often contains few homogeneous instances within a single image,

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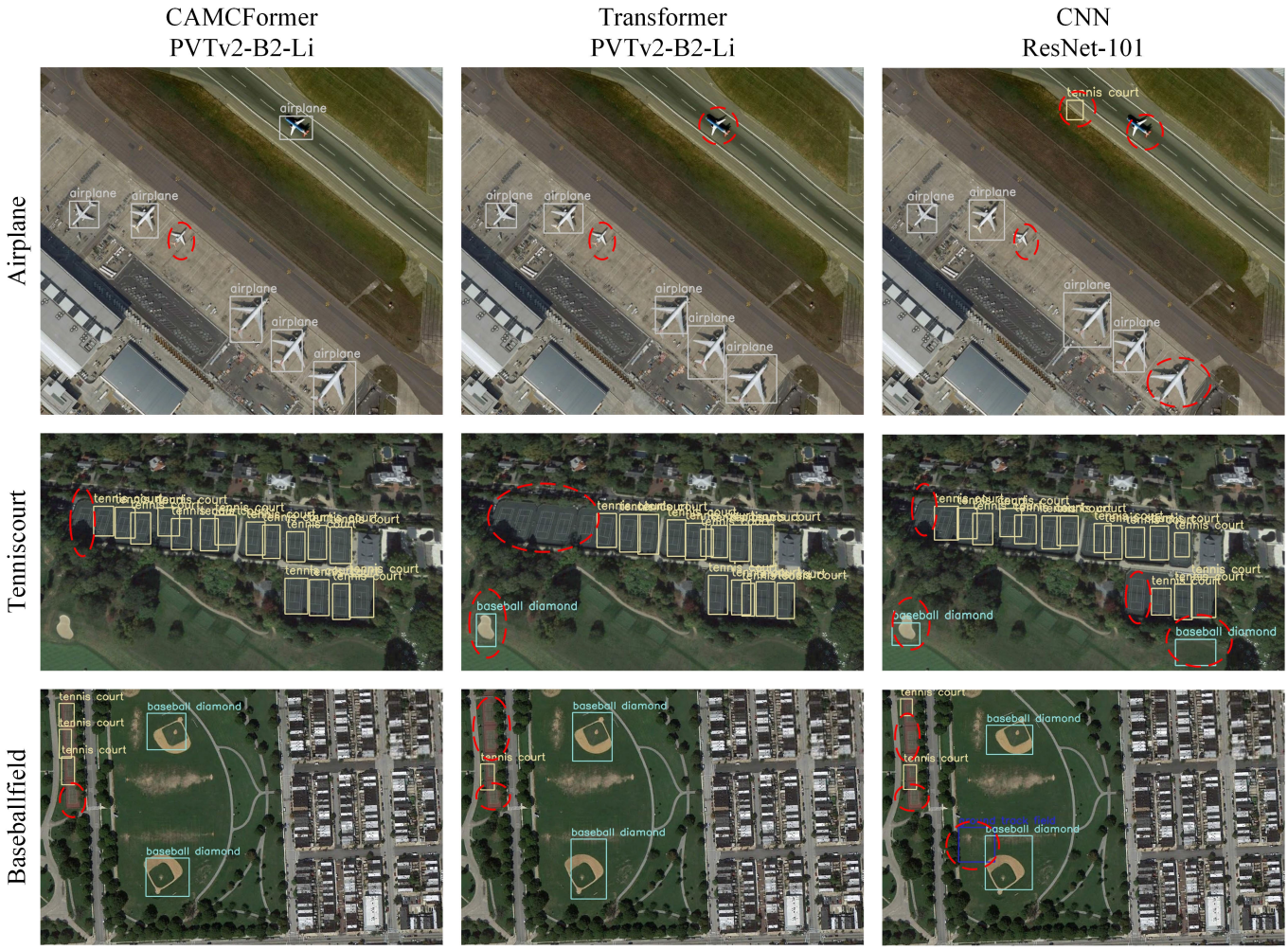


Fig. 7. Visualization of experimental results on three novel classes of the NWPU VHR-10 dataset under 20-shot setting. Only the Bboxs with a confidence greater than 0.5 are visualized. The objects circled by the red ellipse are missed or misclassified.

reducing the effectiveness of long-range modeling and cross-attention mechanisms.

Fig. 8 also illustrates the detection results on five novel classes of the DIOR dataset: airplane, tennis court, train station, baseball field, and windmill, under a 20-shot setting with bounding boxes with confidence scores above 0.5. It is observed that the proposed method achieves the best overall performance. The naive transformer framework (without the LDCAM) with MAHs shows more missed detections and misclassifications. For example, it frequently misidentifies airplanes as windmills, fails to detect windmills, and incorrectly classifies tennis courts as ground track fields. These errors highlight the importance of the LDCAM in enhancing the model's ability to capture fine-grained features and contextual information, which are crucial for accurate object detection. The CNN-based ResNet-101 model performs the worst, with more frequent missed detections and incorrect classifications. For instance, it misidentifies windmills as airplanes, misses the detection of airplanes, incorrectly classifies background regions as tennis courts, and misclassifies baseball fields as ground track fields. These issues suggest that the ResNet-101, while effective in many general object detection tasks, struggles with the specific challenges posed by the novel classes in the DIOR dataset.

D. Ablation Studies

To validate the effectiveness of the proposed CAMCFormer framework, LDCAM, and the MAHs, extensive ablation experiments were conducted on the NWPU VHR-10 dataset.

1) Ablation Study for CAMCFormer Framework: To validate the effectiveness of the proposed CAMCFormer framework, we conducted a comparative analysis with two other state-of-the-art backbones: ResNet-101 and PVTv2-B2-Li. Specifically, we selected the two-branch meta-learning FSOD method, Meta Faster R-CNN [16], a well-established approach in the field. We implemented meta faster R-CNN using the three backbones: ResNet-101, PVTv2-B2-Li, and our proposed CAMCFormer with the LDCAM. As shown in Table VI(a)–(c), the performance of the CAMCFormer framework significantly outperforms both ResNet-101 and PVTv2-B2-Li backbones across various evaluation metrics.

2) Ablation Study for LDCAM: To assess the impact of the LDCAM across four different stages, results are presented in

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TABLE V

QUANTITATIVE EXPERIMENTAL RESULTS OF THE PROPOSED METHOD AND EXISTING SOTA FSOD METHODS ON DIOR DATASET. BOLD FONT INDICATES THE BEST RESULT IN THE GROUP. SYMBOL "-" REPRESENTS UNREPORTED RESULTS IN THE ORIGINAL WORK. THE RESULTS ARE REPORTED BY AVERAGING OVER TEN RUNS

Shot	Methods	Base Classes	Novel Classes					nAP
		bAP	windmill	airplane	tennis court	train station	baseball field	
3	Meta R-CNN [55]	46.3	9.1	15.1	31.8	7.1	5.2	13.7
	FsDetView [56]	36.8	9.1	9.7	22.8	6.9	4.3	10.6
	DAnA [66]	43.8	0.3	1.9	19.4	14.1	19.3	11.0
	FSCE [69]	56.3	9.1	9.1	6.6	9.9	27.3	12.4
	DCFS [67]	-	9.1	11.7	23.6	3.5	37.0	17.0
	MFDC [68]	59.4	11.2	10.1	35.2	7.0	40.1	20.8
	MLII-FSOD [70]	53.1	10.4	19.2	22.5	15.6	36.8	20.9
	FCT [57]	61.9	8.3	15.5	51.8	6.8	49.8	26.5
	CAMCFormer(ours)	**67.1**	**11.6**	**20.6**	**52.3**	**18.1**	**53.9**	**31.3**
5	Meta R-CNN [55]	47.1	9.1	17.0	33.2	8.9	7.2	15.1
	FsDetView [56]	37.8	9.1	22.3	26.2	5.2	2.3	13.0
	DAnA [66]	49.5	2.4	11.4	26.8	12.5	36.5	17.9
	FSCE [69]	56.9	**15.6**	9.1	40.6	10.2	41.9	23.5
	DCFS [67]	-	13.1	15.8	39.5	8.5	43.5	24.1
	MFDC [68]	60.1	14.8	13.4	40.5	10.5	45.5	24.9
	MLII-FSOD [70]	53.3	12.8	21.9	30.2	**23.4**	38.4	25.3
	FCT [57]	63.2	7.9	18.7	53.7	18.6	**53.5**	30.5
	CAMCFormer(ours)	**64.2**	13.3	**28.0**	**57.3**	18.3	52.1	**33.8**
10	Meta R-CNN [55]	46.7	11.4	33.5	32.4	14.0	3.5	18.9
	FsDetView [56]	39.7	11.2	20.4	25.6	7.0	6.1	14.1
	DAnA [66]	50.4	6.5	21.1	27.9	15.9	25.5	19.4
	FSCE [69]	57.6	27.5	9.1	38.8	14.7	47.2	27.5
	DCFS [67]	-	15.9	18.9	48.8	14.4	46.9	29.0
	MFDC [68]	60.2	15.6	17.8	47.7	16.4	49.0	29.3
	MLII-FSOD [70]	54.8	**27.7**	20.5	38.4	26.2	47.8	32.1
	FCT [57]	64.7	10.8	27.0	54.7	25.0	54.5	34.4
	CAMCFormer(ours)	**65.8**	15.7	**33.8**	**58.2**	**30.6**	**54.9**	**38.6**
20	Meta R-CNN [55]	48.1	10.2	34.5	35.4	13.7	8.5	20.5
	FsDetView [56]	41.8	16.2	31.7	25.8	11.1	6.1	18.2
	DAnA [66]	53.5	10.4	14.3	28.6	16.5	38.5	21.7
	FSCE [69]	60.2	34.1	20.1	49.1	13.5	51.1	33.6
	DCFS [67]	-	14.0	19.2	49.7	16.7	49.7	29.9
	MFDC [68]	58.6	16.0	18.3	48.3	16.6	45.9	29.0
	MLII-FSOD [70]	56.1	**35.2**	34.2	39.1	26.5	48.2	36.6
	FCT [57]	64.5	12.3	30.3	55.1	23.0	52.5	34.6
	CAMCFormer(ours)	**68.6**	26.1	**37.7**	**59.1**	**33.2**	**56.4**	**42.5**

TABLE VI

PERFORMANCE COMPARISON WITH DIFFERENT BACKBONES ON 3-SHOT, 5-SHOT, AND 10-SHOT FSOD TASKS, USING RESNET-101, PVTV2-B2-LI, AND CAMCFORMER

No.	Method	Backbone	3-Shot			5-Shot			10-Shot		
			AP	AP50	AP75	AP	AP50	AP75	AP	AP50	AP75
(a)	Meta Faster R-CNN [16]	ResNet-101	8.5	26.1	2.5	11.3	28.6	7.5	16.1	30.5	9.8
(b)		PVTv2-B2-Li	9.6	27.9	7.5	12.8	30.4	9.5	19.1	32.6	13.2
(c)		CAMCFormer	**15.8**	**35.6**	**11.0**	**24.4**	**46.3**	**22.9**	**35.1**	**65.3**	**35.3**

Table VII(d)–(i). It is observed that the best performance is achieved when the LDCAM is applied across all four stages, improving AP, AP50, and AP75 metrics under the 3-shot, 5-shot, and 10-shot settings. When LDCAM is used individually at one stage, the results improve as the stage becomes deeper within the feature extraction backbone, since deeper layers contain richer semantic information. Additionally, it is observed that utilizing LDCAM across the first three feature

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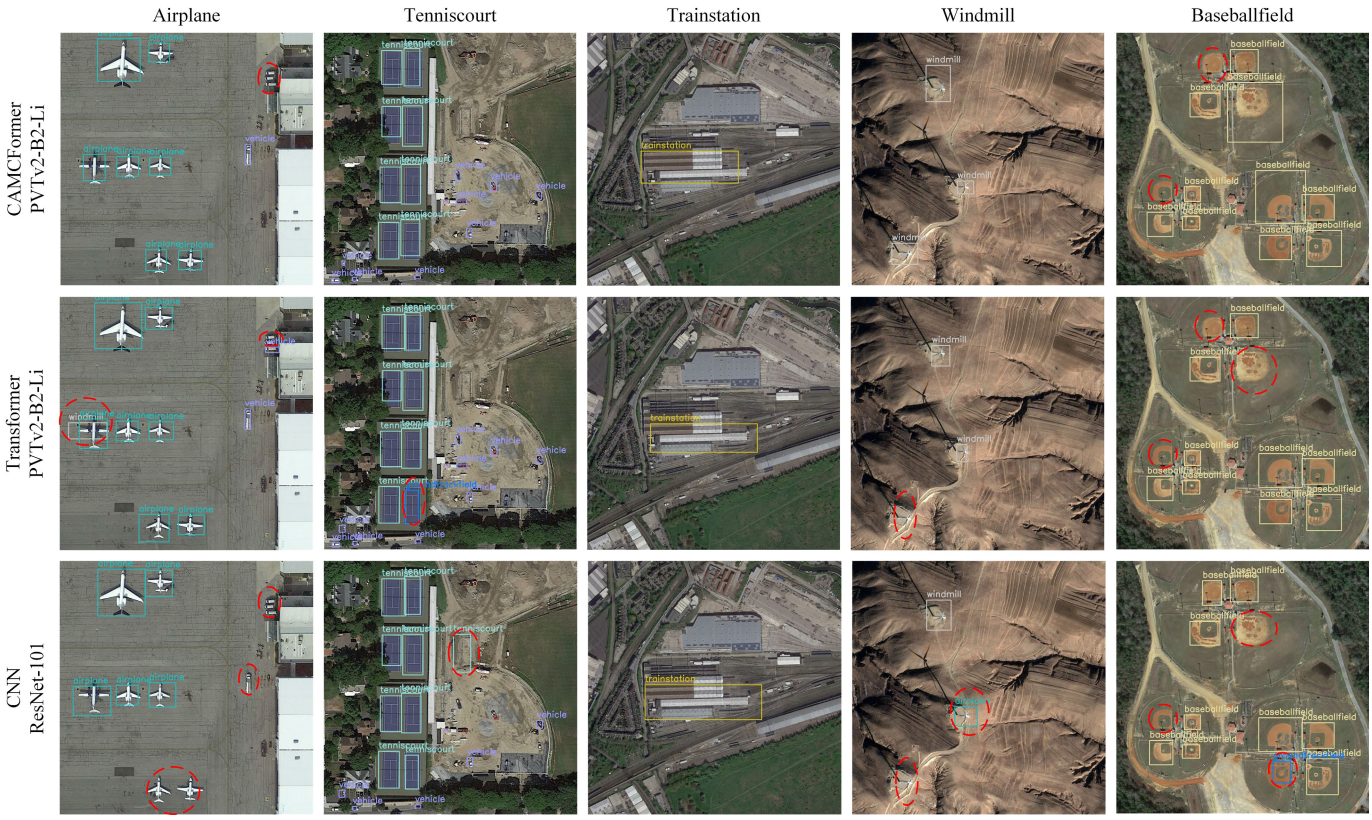


Fig. 8. Visualization of experimental results on five novel classes of the DIOR dataset under 20-shot setting. Only the Bboxs with a confidence greater than 0.5 are visualized. The objects circled by the red ellipse are missed or misclassified.

TABLE VII

ABLATION STUDY OF CAMCFORMER'S PERFORMANCE ON FSOD TASKS (3-SHOT, 5-SHOT, AND 10-SHOT) WITH DIFFERENT STAGE CONFIGURATIONS OF LDCAM USING THE PVTV2-B2-LI BACKBONE

No.	Backbone	LDCAM				3-Shot			5-Shot			10-Shot		
		Stage 1	Stage 2	Stage 3	Stage 4	AP	AP50	AP75	AP	AP50	AP75	AP	AP50	AP75
(d)	PVTv2-B2-Li	✓				16.8	36.1	11.5	17.9	43.8	10.5	29.3	61.4	21.3
(e)	PVTv2-B2-Li		✓			15.6	33.1	11.8	20.1	47.5	11.6	31.2	65.4	24.5
(f)	PVTv2-B2-Li			✓		21.1	45.9	15.0	27.6	55.2	21.0	30.4	64.5	23.1
(g)	PVTv2-B2-Li				✓	23.7	50.5	15.9	27.9	54.3	25.3	33.9	66.8	30.4
(h)	PVTv2-B2-Li	✓	✓	✓		22.0	45.5	20.6	25.9	57.5	20.6	32.5	67.7	26.4
(i)	PVTv2-B2-Li	✓	✓	✓	✓	**29.3**	**56.8**	**24.6**	**30.8**	**61.0**	**29.6**	**39.5**	**75.4**	**37.6**
(j)	PVTv2-B0	✓	✓	✓	✓	8.4	22.3	2.6	13.6	25.5	6.7	13.6	24.6	10.7
(k)	PVTv2-B1	✓	✓	✓	✓	15.9	28.1	5.5	11.3	29.8	13.9	16.4	35.5	19.5
(l)	PVTv2-B2	✓	✓	✓	✓	23.3	52.6	17.6	28.1	50.2	19.5	29.4	53.7	21.1

extraction stages slightly outperforms using it only in the fourth stage. This indicates that integrating multiple modules yields better performance. However, incorporating LDCAM in the detection stage proves crucial, as it models long-range dependencies between proposals and support features, making it highly beneficial for the detector.

Table VII(j)–(l) compares the performance of different PVTv2 backbones. PVTv2-B0 and PVTv2-B1 exhibit significantly lower performance in (j) and (k) compared to PVTv2-B2 (l) due to their limited representational capacity relative to PVTv2-B2-Li. PVTv2-B2 and PVTv2-B2-Li share similar network architectures, leading to comparable performance. However, PVTv2-B2-Li is more computationally efficient and enhances the effectiveness of LDCAM, resulting in slightly better performance in (i) than in (l).

3) Ablation Study for MAHs: To validate the effectiveness of MAHs, various combinations of classification heads were systematically evaluated. As shown in Table VIII, the results indicate optimal performance is achieved when channel-correlation, spatial-correlation, and CADHs are used simultaneously. When evaluating the performance of individual detection heads, channel-correlation and cross-attention demonstrate similar effectiveness, while the spatial-correlation head shows relatively weaker performance. This can be attributed to spatial-correlation's limited assistance in extracting features of different spatial scales

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TABLE VIII
ABLATION STUDY ON EACH CLASSIFICATION HEAD OF MAHS IN OUR MODEL

Multi-Correlation Aided Heads			5-Shot			10-Shot		
CCDH	SCDH	CADH	AP	AP50	AP75	AP	AP50	AP75
✓			24.3	49.1	19.1	27.5	63.8	17.1
	✓		24.1	42.2	16.9	29.0	63.8	23.2
		✓	26.5	49.8	17.1	30.9	65.0	26.6
✓	✓		26.2	52.4	25.4	36.4	68.6	32.4
✓		✓	26.5	53.1	22.6	37.2	70.2	35.3
	✓	✓	28.1	56.5	21.3	35.5	73.0	27.6
✓	✓	✓	**30.8**	**61.0**	**29.6**	**39.5**	**75.4**	**37.6**

Abbreviation of MAHs - CCDH: Channel-Correlation Detection Head; SCDH: Spatial-Correlation Detection Head; CADH: Cross-Attention Detection Head.

TABLE IX
ABLATION STUDY ON BOUNDING BOX REGRESSION OF MAHS WITH DIFFERENT INPUT FEATURES

Regression Head			5-Shot			10-Shot		
F_{cc}	F_{sc}	F_{ca}	AP	AP50	AP75	AP	AP50	AP75
✓			**29.3**	**56.8**	**24.6**	**39.5**	**75.4**	**37.6**
	✓		25.9	54.0	18.4	37.1	70.2	35.3
		✓	27.8	54.9	22.3	38.0	71.4	34.9

for classification. However, spatial-correlation effectively enhances overall detection performance when supplementing channel-correlation and cross-attention.

Due to the diverse and complex visual patterns exhibited by different object categories, utilizing multiple parallel classification heads helps capture a wide range of feature representations across these categories. However, the regression task, which predicts the coordinates of bounding boxes, is a continuous problem with lower complexity than the classification task, which involves identifying distinct categories. A single regression head is computationally more efficient and sufficient to learn spatial relationships and predict the precise location of objects within an image. Therefore, a single head is used for the regression task.

To analyze the impact of different feature correlations on the regression task, various correlation features (channel-correlation features $\boldsymbol{F}_{\mathrm{cc}}$, spatial-correlation features $\boldsymbol{F}_{\mathrm{sc}}$, and cross-attention features $\boldsymbol{F}_{\mathrm{ca}}$) were used separately as input to the regression task. As shown in Table IX, channel-correlation features are more suitable for the regression task so that $\boldsymbol{F}_{\mathrm{cc}}$ are adapted in the proposed method.

E. Analysis

1) Analysis of Sensitivity of Key Hyperparameters: To guide the optimal model configuration and improve generalization performance, we conducted a sensitivity analysis of the model's key hyperparameters: the number of attention heads, decay rate, and learning rate. The experimental results are presented in Table X. The number of attention heads refers to the specific configuration for the four stages, each with its number of attention heads. The decay rate specifies the step schedule, starting to decay at either 2000 or 3000 steps and continuing to a maximum of 4000 steps, with decay factors of 0.1 or 0.2. The learning rate represents the base learning rate used in training. The results indicate that the optimal configuration occurs when the number of attention heads is set to [1, 2, 5, 8], the decay rate starts at step 3000 or 4000 with a factor of 0.1, and the learning rate is initialized to 0.0002. This configuration achieved the best overall performance. "-" indicates that no results were obtained.

TABLE X
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS

Hyperparameters	Value	Shot			
		3	5	10	20
Number of attention heads	[1, 2, 4, 4]	37.3	42.1	62.4	75.1
	[1, 2, 5, 8]	**54.2**	**60.0**	**77.2**	**80.1**
	[1, 2, 8, 8]	40.7	54.0	65.5	69.0
Decay rate	[3000, 4000], 0.1	**54.2**	**60.0**	**77.2**	**80.1**
	[3000, 4000], 0.2	39.3	56.0	59.8	71.9
	[2000, 4000], 0.2	34.6	40.6	57.0	71.5
Learning rate	0.002	-	-	-	-
	0.0002	**54.2**	**60.0**	**77.2**	**80.1**
	0.0001	46.4	55.2	67.4	78.7

2) Analysis of Model Architectures: To validate the effectiveness of the proposed framework, we visualized the attention map of different network architectures, including the CNN-based ResNet-101, the naive Transformer PVTv2-B2-Li, and the proposed CAMCFormer. For testing, three novel classes (airplane, baseball diamond, and tennis court) were selected from the NWPU VHR-10 dataset. One test image was randomly chosen for each class and fed into the trained networks. The heatmaps were extracted from the third feature extraction stage in the backbone networks, enabling us to observe the attention distribution. All networks were trained under the ten-shot setting, ensuring a consistent evaluation framework.

As shown in Fig. 9, the proposed CAMCFormer demonstrates a significantly better perception of the novel classes than ResNet-101 and naive Transformer-based architecture. ResNet-101 fails to capture the attention regions for the baseball diamond and tennis court classes. The naive Transformer model detects the novel classes in airplane and baseball diamonds but misses some relevant areas. In contrast, the CAMCFormer successfully perceives all the novel classes, such as airplane, baseball diamond, and tennis court, with exact attention regions, particularly for the tennis court.

By visualizing the attention weights between the query image and its corresponding support image, as shown in Fig. 9, it is evident that the attention in the support image focuses on its distinctive and discernible regions. These regions serve to emphasize the corresponding target areas in the query image. This alignment effectively highlights the relevant features in the query image that correspond to the support image, demonstrating the model's ability to leverage contextual information for accurate detection. This visualization underscores

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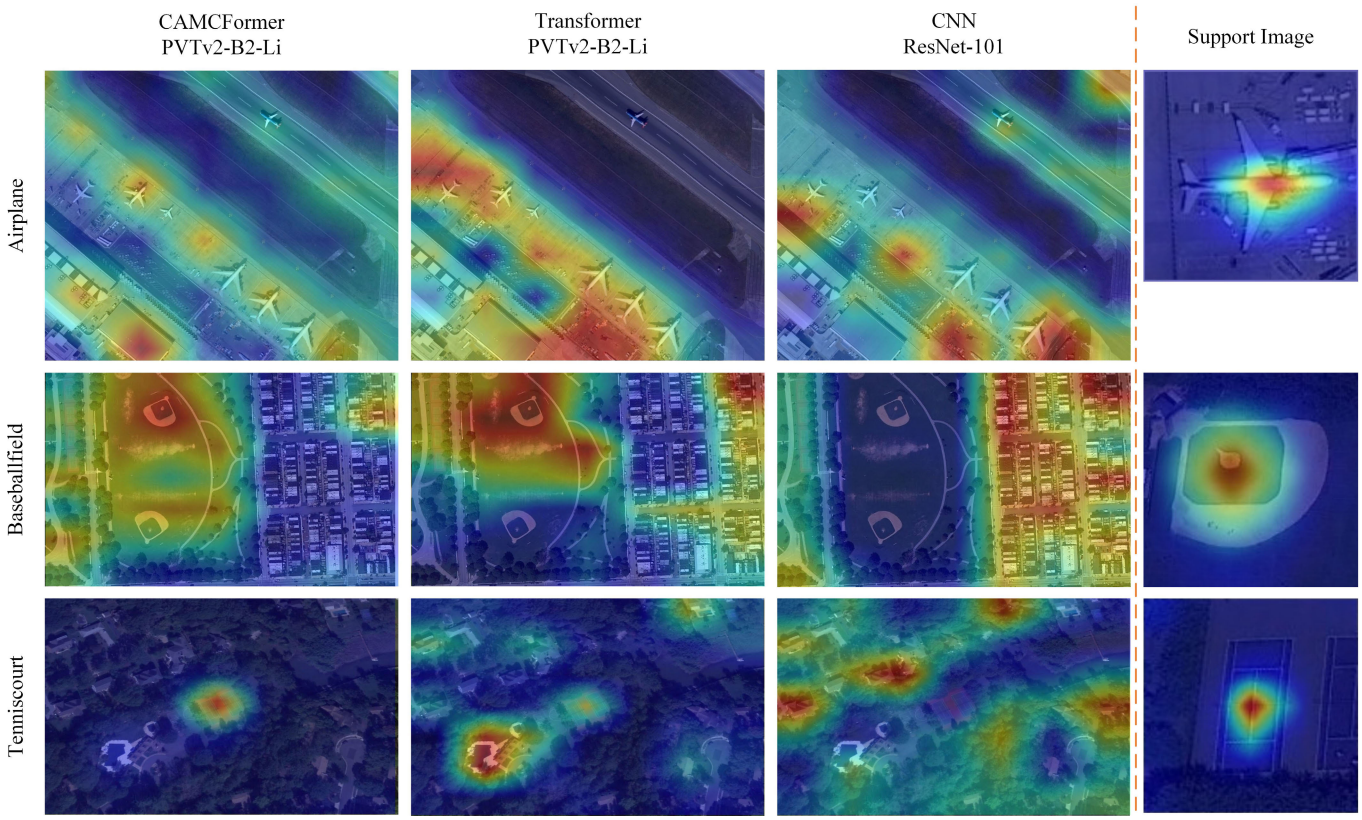


Fig. 9. Comparisons of the different models' visual attention on novel classes under 10-shot condition and visualization of the attention weights between the query image and its corresponding support image.



Fig. 10. Visualization of CAMCFormer's results under noisy conditions.

the strength of the proposed attention mechanism in capturing relational dependencies between the query and support images.

3) Analysis Under Noisy Conditions: To validate the model's robustness in real-world remote sensing scenarios, we selected images from the DIOR test dataset with challenges such as cloud occlusion, noise blur, low illumination, and shadows, and conducted 10-shot detection. The results, visualized in the Fig. 10, show that the model successfully detects the target locations in most cases, demonstrating its effectiveness and robustness.

IV. Conclusion

The CAMCFormer FSOD framework is proposed for RSIs to model global relational dependencies, addressing the challenges of detecting objects with limited labeled samples. This framework introduces a LDCAM, which effectively captures dependencies between distant elements in both query and support images during feature extraction, allowing for the comprehensive modeling of global relationships. By facilitating the interaction of contextual information across images, the LDCAM enhances feature representations and eliminates the need for additional alignment and fusion modules that are commonly required in traditional methods. Additionally, the framework incorporates MAHs, which target various relational aspects, such as channel correlation, multispatial scale correlation, and cross-attention union correlation. These specialized heads improve the framework's ability to capture complex relationships across features, thus enhancing the robustness and accuracy of predictions in both classification and localization tasks. Through extensive experiments on standard remote sensing datasets, the proposed CAMCFormer framework demonstrates superior performance compared to several state-of-the-art detectors, validating its potential as an effective solution for FSOD in remote sensing applications.

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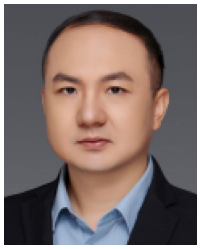
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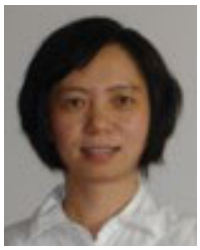
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